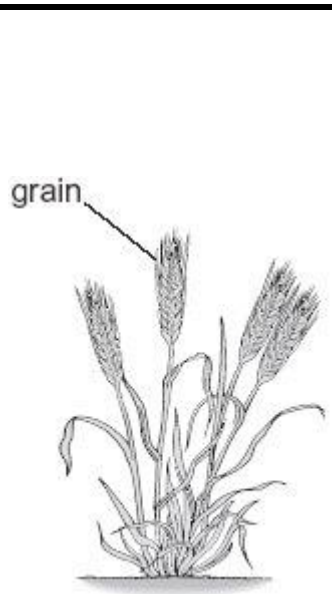
Q1. (a) The drawings below show an old and a modern variety of wheat plant.

	 **old variety**	 **modern variety**
average mass of grain produced per m² (kg)	0.5	0.8
average length of stalk (cm)	145	78

Glucose produced by the wheat plants is used:

- to provide energy for growth
- to make cell walls
- to make starch which is stored in the grain.

Give **one** reason why modern wheat plants with short stalks can store more starch in the grain. Use the drawings and information.

.....

.....

1 mark

(b) A plant breeder wants to use selective breeding to produce corn with short stalks and a high mass of grain. He could use the following varieties of corn:

variety A	**variety B**	**variety C**
long stalks	short stalks	long stalks
high mass of grain	low mass of grain	low mass of grain

(i) What would the plant breeder need to do to make sure he always produced corn with short stalks and a high mass of grain?
Describe the three steps the breeder would use.

.....

.....

.....

.....

.....

.....

3 marks

(ii) Suggest **one** other characteristic that farmers might like corn plants to have to increase the amount of corn produced.

.....

1 mark
maximum 5 marks

Q1.

(a) any **one** from

- less cellulose is needed or fewer cell walls produced **or** less glucose changed to cellulose

  accept 'less stalk is made'

- less glucose **or** energy needed **or** used for growth

  accept 'less glucose used in respiration'
  'less starch is used for growth' is insufficient
  accept 'it does not have to grow as much'

- more glucose is available to be changed into starch

  accept 'more glucose goes to the grain'
  *do **not** accept 'the glucose gets to the grain quicker'*
  accept 'blown over less easily or less frequently'

1 (L7)

(b) (i) any **two** from

- cross plants with short stalks and low grain mass with plants with long stalks and high grain mass

  accept 'breed from varieties A and B'
  'mix varieties A and B' is insufficient

- collect **or** plant the seeds

- choose offspring with shortest stalks and which produce a high mass of grain

  accept 'choose the best'

2 (L7)

- repeat with offspring or continue the whole process

  a three mark answer must include two of the first three marking points and a reference to repeating or continuing the process

1 (L7)

(ii) any **one** from

- disease resistance

- pest resistance

  'increased amount of grain'
  ***or** 'number of ears of corn' are insufficient*
  'more stalks' is insufficient
  accept 'corn' for 'wheat'

- drought tolerance

- frost resistance

- resistance to herbicides **or** weed killers

- resistance to wind

  *accept 'strong stalk' **or** 'long roots'*
  accept 'hardy'
  *accept 'big **or** long leaves **or** lots of leaves'*
  accept 'faster growing'

1 (L7)

[5]